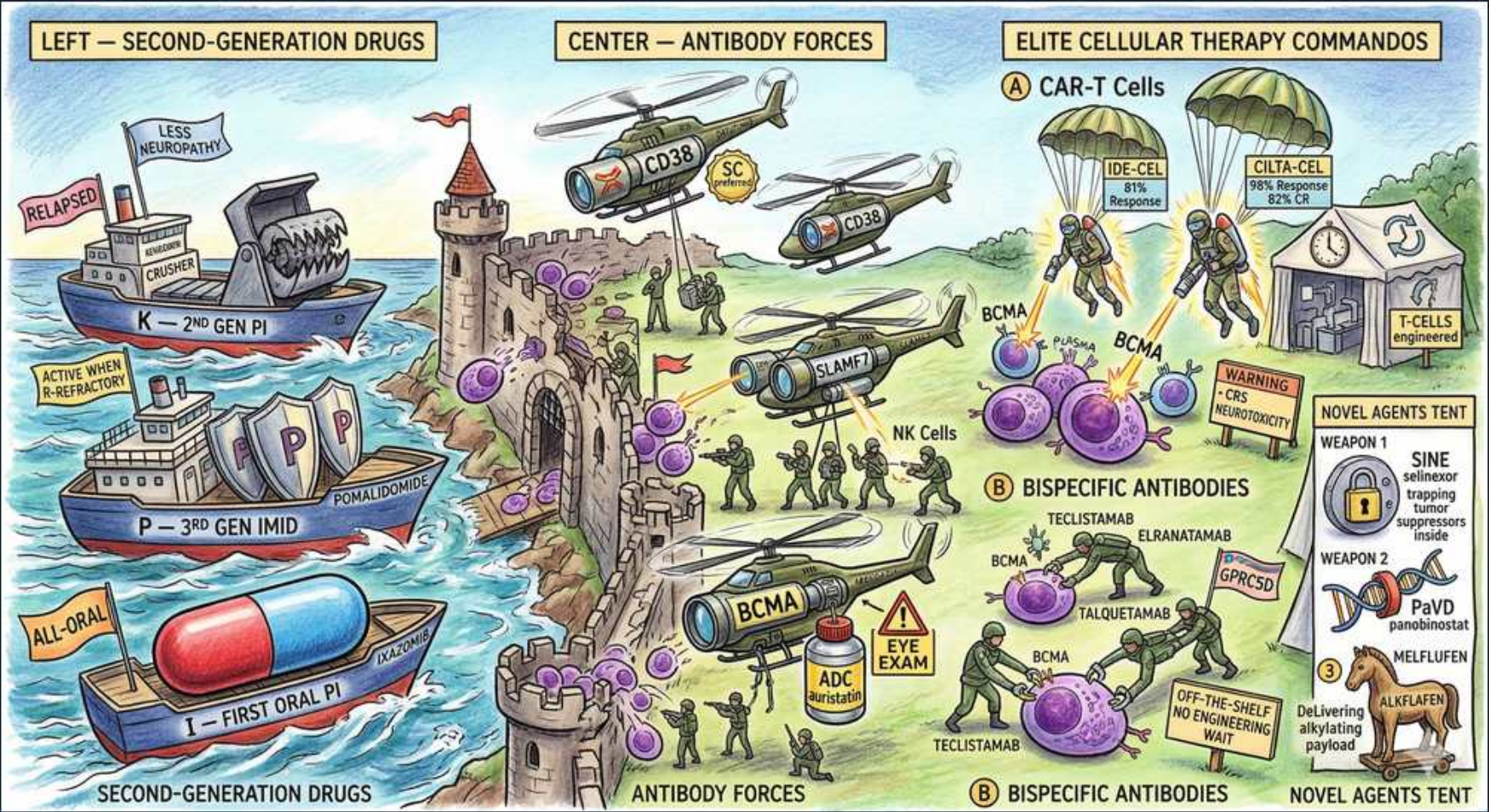


Relapsed/Refractory Myeloma — CAR-T, Bispecifics & Novel Agents



1. 2nd-gen PI — carfilzomib (relapsed)

WHAT YOU SEE

A 'SECOND-GENERATION DRUGS' warship 'K — 2nd-gen PI' with shark teeth, labeled 'relapsed'.

WHAT IT ENCODES

Carfilzomib is a second-generation proteasome inhibitor used in relapsed/refractory disease (e.g., KRd, Kd, Dara-Kd). It is more potent than bortezomib and lacks significant neuropathy, but is CARDIOTOXIC (heart failure, hypertension, dyspnea) and can cause thrombotic microangiopathy.

BOARD TRAP

Carfilzomib's hallmark toxicity is CARDIOVASCULAR (HF, hypertension, TMA), not neuropathy — screen cardiac risk and monitor BP; this distinguishes it from bortezomib.

2. 3rd-gen IMiD — pomalidomide

WHAT YOU SEE

A ship 'P — 3rd-gen IMiD', noted 'active when len-refractory'.

WHAT IT ENCODES

Pomalidomide is the third-generation IMiD active even in lenalidomide-refractory disease, used in combinations like Pom-dex, Dara-Pd, Isa-Pd, and Elo-Pd. Same class toxicities: myelosuppression, VTE (needs prophylaxis), and teratogenicity.

BOARD TRAP

Pomalidomide retains activity AFTER lenalidomide failure — it is the go-to IMiD in len-refractory relapse, still requiring VTE prophylaxis and pregnancy prevention.

3. First oral PI — ixazomib (all-oral)

WHAT YOU SEE

A red-and-white capsule ship 'I — FIRST ORAL PI', labeled 'ALL-ORAL'.

WHAT IT ENCODES

Ixazomib is the first oral proteasome inhibitor, enabling all-oral triplets (IRd) — convenient for relapsed patients wanting to avoid infusions, with minimal neuropathy. Modest potency relative to IV PIs but valued for tolerability and convenience.

BOARD TRAP

Ixazomib's role is convenience (oral, low neuropathy), not superior efficacy — choose it for tolerability/logistics, not as the most potent PI.

4. BCMA CAR-T (ide-cel / cilta-cel)

WHAT YOU SEE

An 'ELITE CELLULAR THERAPY COMMANDOS' panel: ide-cel (~73% response) and cilta-cel (~98% response, ~82% CR) paratroopers targeting BCMA.

WHAT IT ENCODES

Anti-BCMA CAR-T cells (idecabtagene vicleucel and ciltacabtagene autoleucel) are autologous engineered T cells with deep, durable responses in triple-class-refractory myeloma (cilta-cel ~98% ORR). Toxicities: cytokine release syndrome (CRS), ICANS/neurotoxicity, prolonged cytopenias, infections.

BOARD TRAP

CAR-T (and bispecifics) target BCMA — and their signature acute toxicities are CRS and neurotoxicity (ICANS), managed with tocilizumab and steroids; cilta-cel also has delayed neurotoxicity (parkinsonism).

5. BCMA bispecific — teclistamab

WHAT YOU SEE

A 'BISPECIFIC ANTIBODIES' squad: teclistamab/elranatamab/linvoseltamab engaging BCMA, 'off-the-shelf, no engineering, no wait'.

WHAT IT ENCODES

Bispecific T-cell engagers (teclistamab, elranatamab, linvoseltamab) bind BCMA on plasma cells and CD3 on T cells, redirecting cytotoxicity. They are OFF-THE-SHELF (no manufacturing wait, unlike CAR-T) but given continuously. Risks: CRS, neurotoxicity, hypogammaglobulinemia, and serious infections (need IVIG, PJP/HZ prophylaxis).

BOARD TRAP

Bispecifics are off-the-shelf (immediate, no apheresis/manufacturing) vs CAR-T's manufacturing delay — but their continuous dosing brings high infection risk requiring IVIG and antimicrobial prophylaxis.

6. Non-BCMA bispecific — talquetamab (GPRC5D)

WHAT YOU SEE

A bispecific squad member 'TALQUETAMAB' targeting 'GPRC5D'.

WHAT IT ENCODES

Talquetamab is a bispecific targeting GPRC5D (not BCMA), useful after BCMA-directed failure. Distinctive on-target toxicities: dysgeusia/taste changes, skin/nail changes, and weight loss, plus CRS. Provides a non-BCMA option to overcome BCMA antigen loss.

BOARD TRAP

Talquetamab targets GPRC5D (alternative antigen) — its unique side effects are taste loss, skin and nail toxicity; it works when BCMA-directed therapy fails due to antigen escape.

7. Anti-CD38 / SLAMF7 antibody forces

WHAT YOU SEE

'ANTIBODY FORCES' helicopters labeled CD38 and SLAMF7 over a castle.

WHAT IT ENCODES

Naked monoclonal antibodies: daratumumab/isatuximab (anti-CD38) and elotuzumab (anti-SLAMF7). Elotuzumab works ONLY in combination (with len-dex or pom-dex) — it has minimal single-agent activity because it acts largely by NK-cell activation. CD38 antibodies anchor many relapse regimens.

BOARD TRAP

Elotuzumab (anti-SLAMF7) has essentially NO single-agent activity — it must be combined with an IMiD; don't expect monotherapy benefit, unlike daratumumab.

8. ADC — belantamab (eye exam)

WHAT YOU SEE

An 'ADC belantamab' unit with an 'EYE EXAM' warning sign.

WHAT IT ENCODES

Belantamab mafodotin is an anti-BCMA antibody-drug conjugate delivering a microtubule toxin (MMAF). Its signature toxicity is OCULAR — keratopathy (corneal microcyst-like epithelial changes) causing blurred vision and dry eyes, requiring scheduled ophthalmology exams and a REMS program.

BOARD TRAP

Belantamab mafodotin's dose-limiting toxicity is KERATOPATHY (eye) — mandatory ophthalmologic monitoring; this corneal toxicity is its defining board association.

9. NK cells (immune effectors)

WHAT YOU SEE

Foot-soldiers labeled 'NK CELLS' supporting the antibody forces.

WHAT IT ENCODES

Many antibodies (anti-CD38, elotuzumab) rely on NK-cell-mediated ADCC. Lenalidomide enhances NK/T-cell function, explaining synergy in IMiD + antibody combinations. Understanding the NK effector mechanism explains why elotuzumab needs an IMiD partner.

BOARD TRAP

Antibody efficacy depends on intact NK/immune effector function — IMiDs boost it, which is why antibody + IMiD combinations outperform either alone.

10. Selinexor (XPO1 / SINE)

WHAT YOU SEE

A 'NOVEL AGENTS TENT — Weapon 1: SINE selinexor', a tumor-suppressor-trapping warhead.

WHAT IT ENCODES

Selinexor is a first-in-class selective inhibitor of nuclear export (SINE) blocking XPO1, trapping tumor-suppressor proteins (p53, etc.) in the nucleus to trigger apoptosis. Used in heavily pretreated/penta-refractory disease. Notable toxicities: nausea, anorexia, weight loss, thrombocytopenia, hyponatremia.

BOARD TRAP

Selinexor blocks XPO1/nuclear export (traps tumor suppressors in the nucleus) — its limiting toxicities are GI (nausea/weight loss), low platelets, and hyponatremia.

11. Venetoclax for t(11;14)

WHAT YOU SEE

A 'Weapon 2: PoVD / venetoclax' note in the novel agents tent.

WHAT IT ENCODES

Venetoclax is a BCL-2 inhibitor with striking activity specifically in t(11;14) (and BCL2-high) myeloma — a biomarker-driven therapy. It is NOT broadly effective in unselected myeloma (a trial showed excess mortality without the t(11;14) biomarker).

BOARD TRAP

Venetoclax in myeloma is reserved for t(11;14)-positive disease — used in unselected patients it can INCREASE mortality; t(11;14) is the predictive biomarker.

12. Melflufen (peptide-drug conjugate)

WHAT YOU SEE

A 'Weapon 3: melflufen', an alkylating peptide-drug conjugate delivering payload.

WHAT IT ENCODES

Melflufen (melphalan flufenamide) is a peptide-drug conjugate that is cleaved by aminopeptidases enriched in myeloma cells to release an alkylator intracellularly, used in heavily pretreated relapse. Its role is limited and overshadowed by immunotherapies; mainly a mechanism to recognize.

BOARD TRAP

Melflufen is a lipophilic peptide-conjugated alkylator activated intracellularly by aminopeptidases — recognize the mechanism; it is a late-line, niche agent.

13. CAR-T vs bispecific — wait time

WHAT YOU SEE

Contrast of CAR-T (needs apheresis/manufacturing) vs bispecific 'no wait, off-the-shelf'.

WHAT IT ENCODES

Key decision point: CAR-T gives the deepest one-time response but needs leukapheresis and weeks of manufacturing (a problem for rapidly progressing disease), whereas bispecifics are immediately available off-the-shelf but require ongoing dosing with cumulative infection risk.

BOARD TRAP

For a rapidly progressing relapse you may favor an off-the-shelf bispecific (no manufacturing delay) over CAR-T — the trade-off is depth/one-time vs immediacy/continuous dosing.

14. BCMA antigen escape

WHAT YOU SEE

Repeated 'BCMA' targeting across CAR-T, bispecifics, and ADC.

WHAT IT ENCODES

BCMA (B-cell maturation antigen, TNFRSF17) is the dominant target across CAR-T, bispecifics, and ADCs. Sequencing multiple BCMA-directed therapies can fail via antigen loss/escape, which is why GPRC5D (talquetamab) and FcRH5-directed agents provide alternative targets after BCMA failure.

BOARD TRAP

Relapse after BCMA-directed therapy may reflect antigen escape — switch to a non-BCMA target (GPRC5D/talquetamab) rather than another BCMA agent.

15. Triple-class-refractory definition

WHAT YOU SEE

Sequential second-gen ships implying exhaustion of PI + IMiD + anti-CD38 classes.

WHAT IT ENCODES

'Triple-class refractory' myeloma is refractory to a proteasome inhibitor, an IMiD, AND an anti-CD38 antibody — the population where CAR-T, bispecifics, selinexor, and novel agents are deployed. Defining this status drives access to immunotherapies.

BOARD TRAP

The phrase 'triple-class refractory' (PI + IMiD + anti-CD38) defines who needs the novel immunotherapies/CAR-T — recognize it as the trigger for these advanced options.